

Vertical dilations



1 Consider the expression $\cos \theta$.

a Complete the table of values for different values of θ .

θ	0	60	90	120	180	240	270	300	360
$\cos \theta$									

b Sketch a graph of the function $y = \cos \theta$.

c State the maximum value of $\cos \theta$.

d State the minimum value of $\cos \theta$.

e State the range of values of $4 \cos \theta$.

2 Consider the functions $y = \sin x$ and $y = \cos x$.

a State the domain of both functions.

b State the range of both functions.

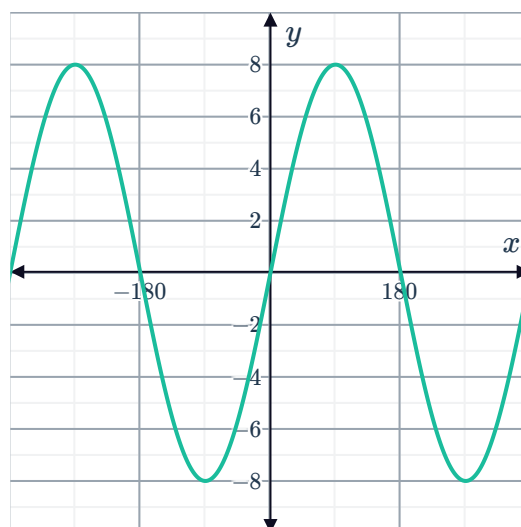
3 Consider the function $y = -2 \cos x$.

a State the domain of the function.

b State the range of the function.

4 Consider the given graph of a function of the form $f(x) = A \sin x$:

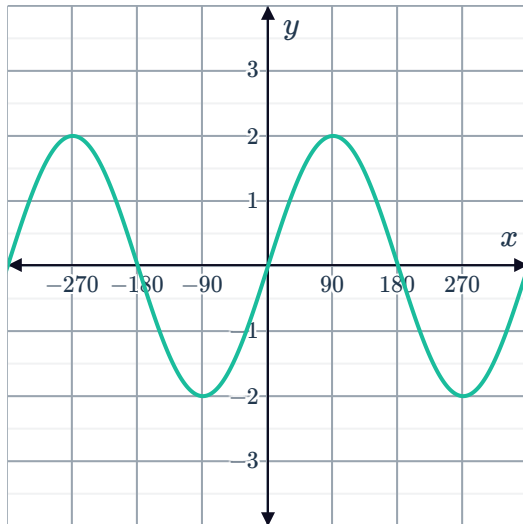
State the amplitude of the function.



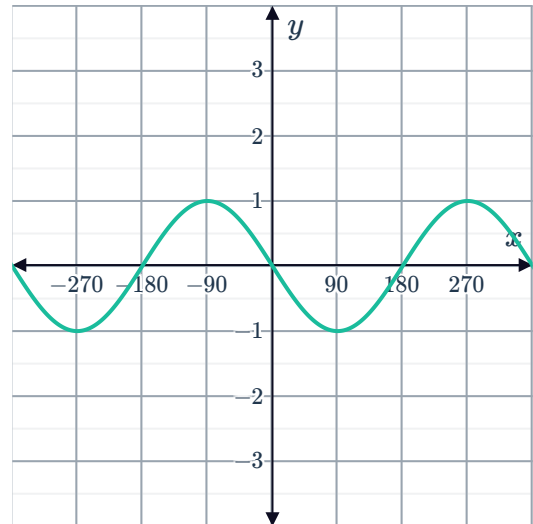
- 5 State the equation of each of the functions plotted below given that they are of the form $y = a \sin x$ or $y = a \cos x$:



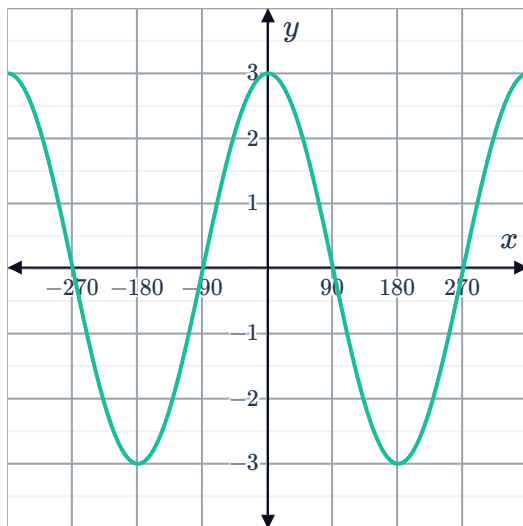
a



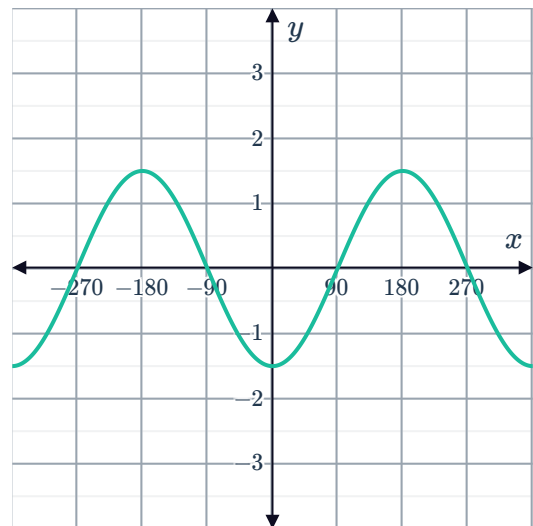
b



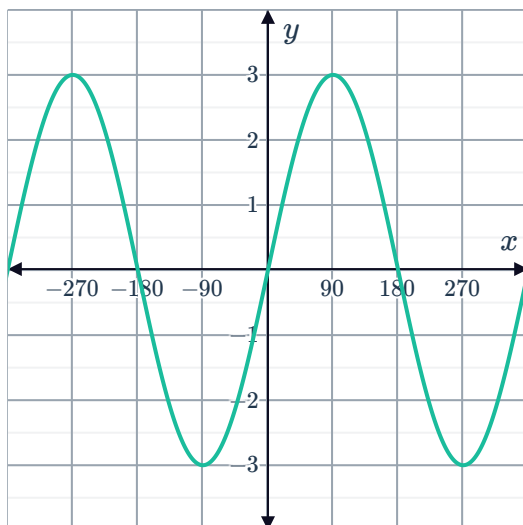
c



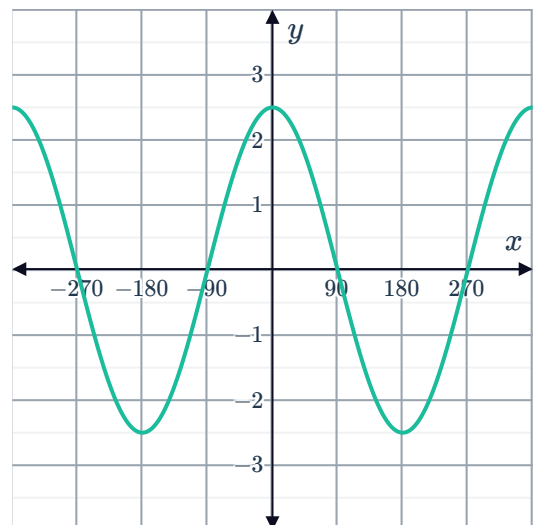
d



e



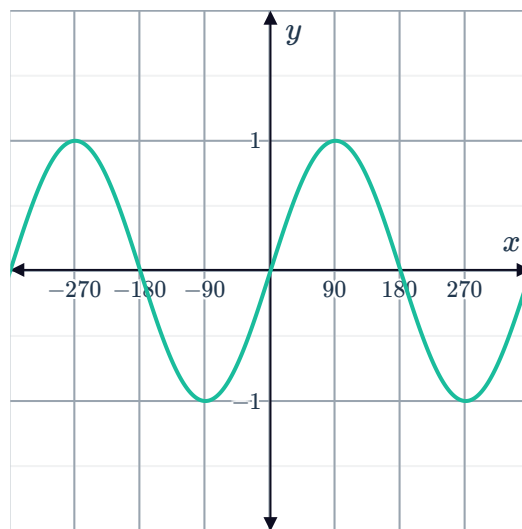
f



- 6 Consider the graph of the function $y = \sin x$ for $0 \leq x < 360$:



At which value of x in the given domain would $y = -\sin x$ have a maximum value?

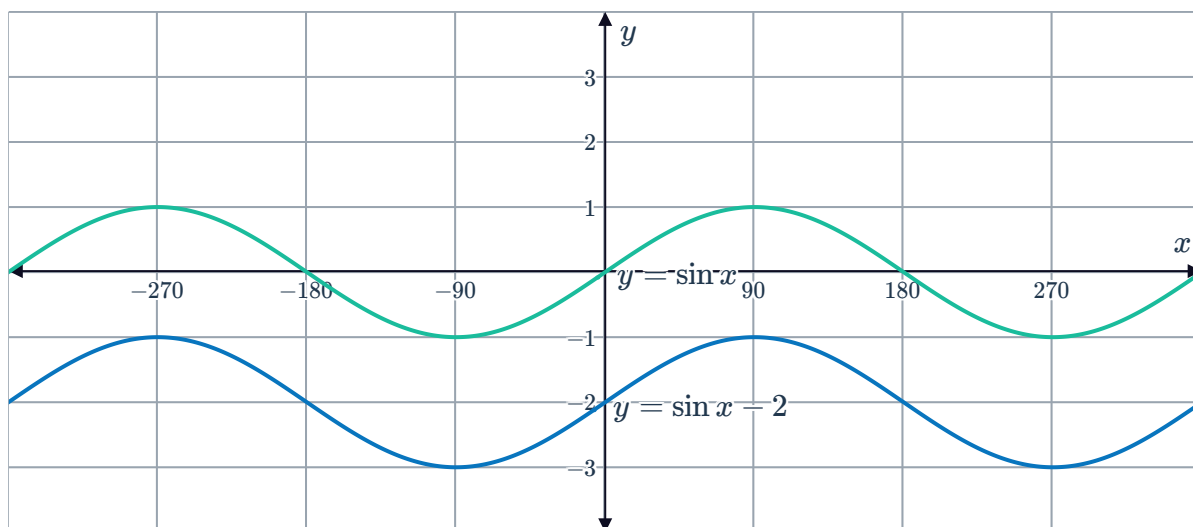


- 7 Consider the function $f(x) = -8 \sin x$, where $0 \leq x \leq 180$.
- State the amplitude of the function.
 - Find the value of $f(180)$.
 - Find the minimum value of the function.
- 8 The function $y = k \sin x$ has a maximum value of 5. Find the value of k , where $k > 0$.
- 9 A sine function has the form $y = c \sin x$, a range of $[-2, 2]$ and a minimum at 90 . Find an expression for y .
- 10 For each of following functions:
- State the amplitude.
 - Sketch a graph of the function.
- $y = 3 \sin x$.
 - $y = 4 \cos x$
 - $y = \frac{4}{3} \cos x$
 - $y = -5 \cos x$
 - $y = \frac{5}{4} \sin x$
 - $y = -4 \sin x$
- 11 Consider the function $y = -3 \cos x$.
- State the maximum value of the function.
 - State the minimum value of the function.
 - State the amplitude of the function.
 - Describe the two transformations required to obtain the graph of $y = -3 \cos x$ from the graph of $y = \cos x$.



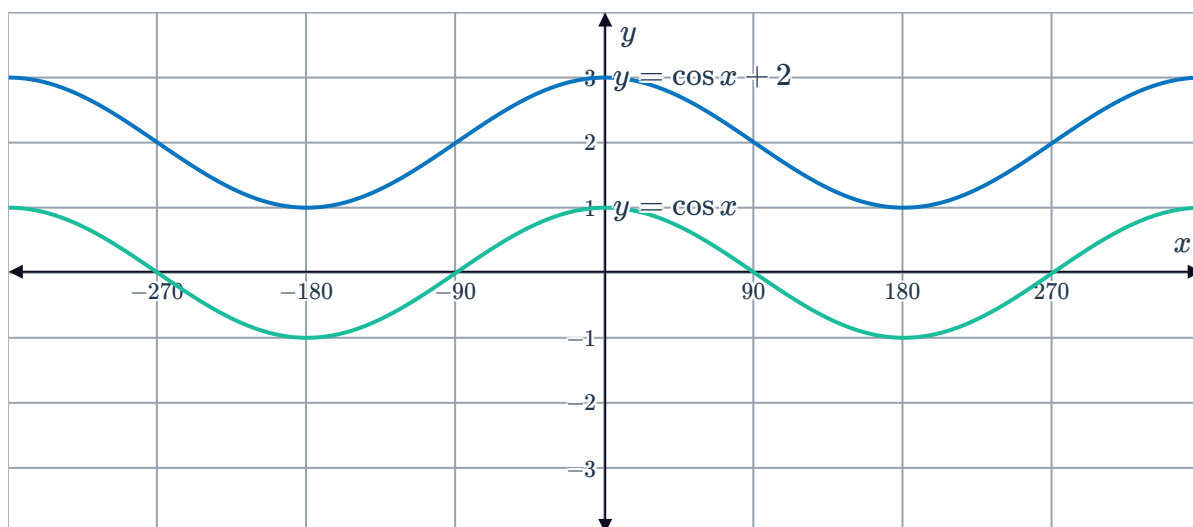
Vertical translations

12 Consider the two graphs $y = \sin x$ and $y = \sin x - 2$ below:



Describe the transformation required to obtain the graph of $y = \sin x - 2$ from $y = \sin x$.

13 Consider the two graphs $y = \cos x$ and $y = \cos x + 2$ below:

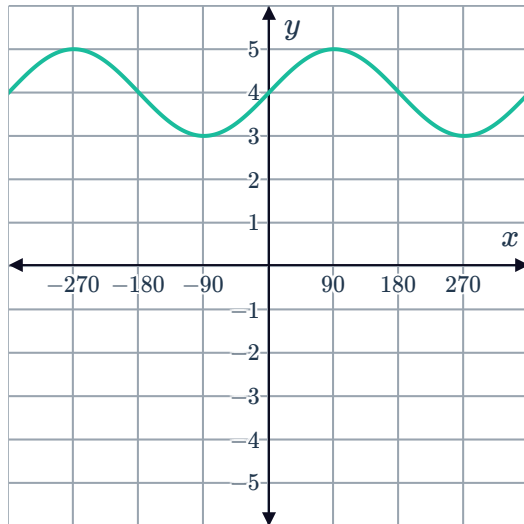


Describe the transformation required to obtain the graph of $y = \cos x + 2$ from $y = \cos x$.

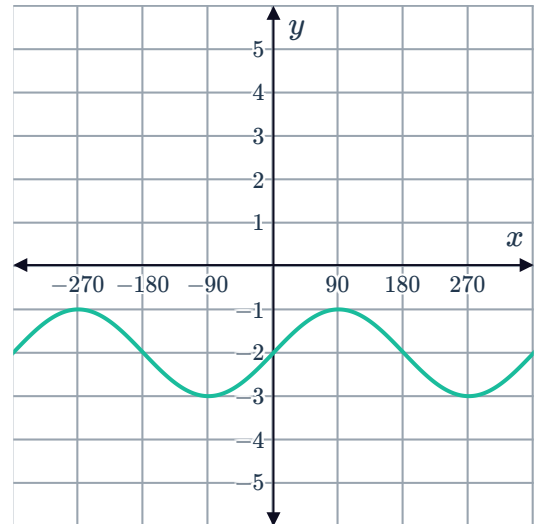
14 Describe the transformation required to obtain the graph of $y = \sin x + 4$ from $y = \sin x$.

- 15 State the equation of each of the functions graphed below given that they are of the form $y = \sin x + k$:

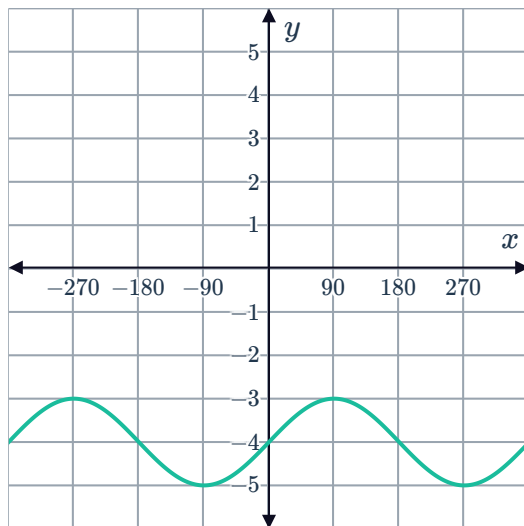
a



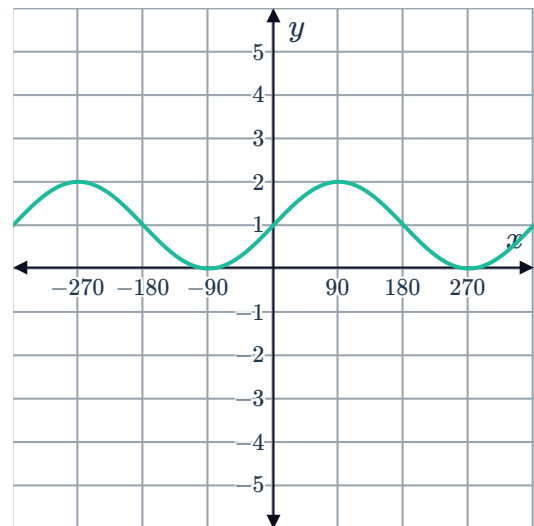
b



c

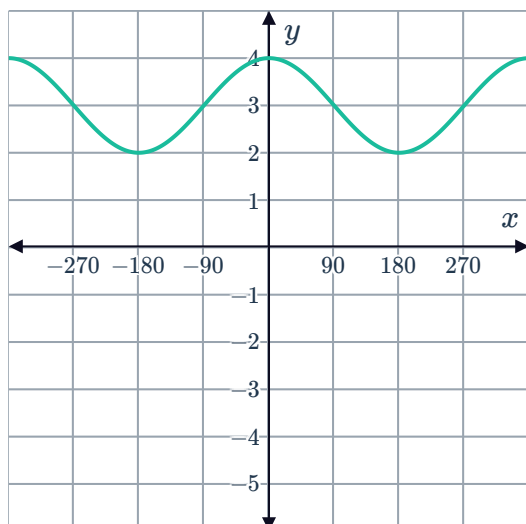


d

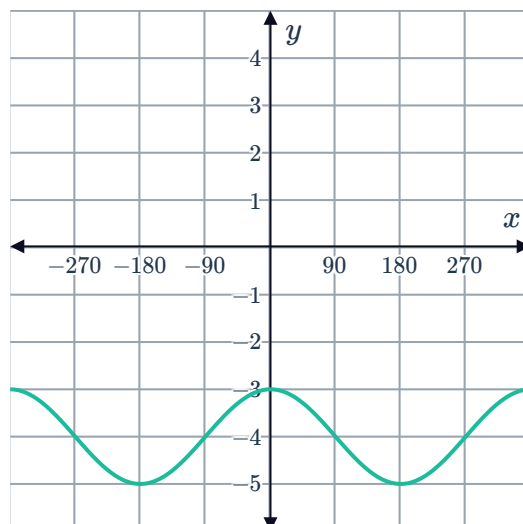


- 16 State the equation of each of the functions graphed below given that they are of the form $y = \cos x + k$:

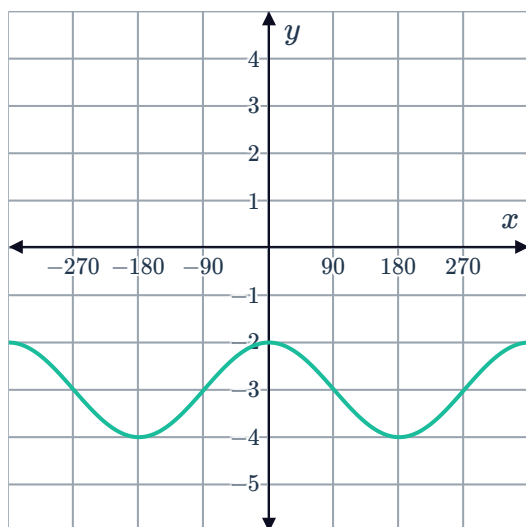
a



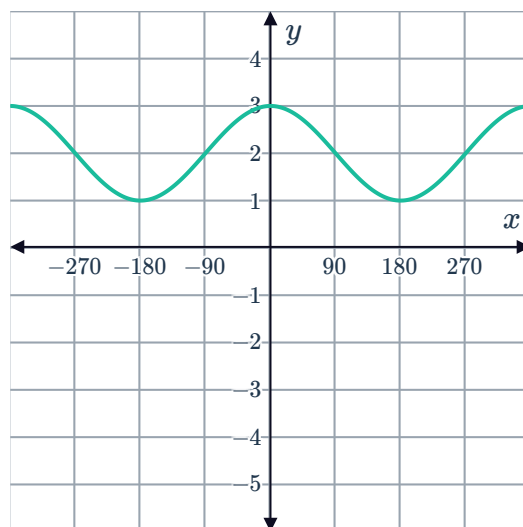
b



c



d



- 17 Describe the two transformations required to turn the graph of $y = \cos x$ into the graph of $y = -\cos x + 3$.

- 18 State the equation of the line that the function $y = \sin x + 2$ oscillates about.

- 19 Consider the function $y = \cos x + 4$.

- Describe the transformation required to obtain the graph of $y = \cos x + 4$ from $y = \cos x$.
- State the period of the function.
- Find the maximum value of the function.
- Find the minimum value of the function.

20 For each of the functions below:



- i State the period of the function.
- ii State the amplitude of the function.
- iii Find the maximum value of the function.
- iv Find the minimum value of the function.
- v Sketch a graph of the function for $-360 \leq x \leq 360$.

a $y = \sin x + 1$ **b** $y = \cos x - 3$ **c** $y = \sin x - 2$ **d** $y = 3 \sin x + 2$
e $y = 3 \cos x - 3$ **f** $y = 2 \sin x - 3$

21 For each of the functions below:

- i Find the value of y when $x = 90$.
- ii State the amplitude of of the function.
- iii Find the maximum value of the function.
- iv Find the minimum value of the function.
- v Describe the transformations required to obtain the graph of the function from $y = \cos x$.
- vi Sketch a graph of the function for $0 \leq x \leq 360$.

a $y = 2 \cos x + 3$ **b** $y = 2 \cos x - 3$ **c** $y = -4 \cos x + 2$ **d** $y = \frac{1}{3} \sin x + 2$

22 The function $y = \cos x + 5$ is translated 4 units up.

- a** State the equation of the new function after the translation.
- b** Find the maximum value of the new function.

23 For each of the functions below:

- i Find the maximum value of the function.
- ii Find the minimum value of the function.

a $y = 2 - 3 \sin x$ **b** $y = 4 - 3 \sin x$ **c** $y = (\sin x)^2 + 1$ **d** $y = \frac{4}{3 \cos x + 5}$

24 A sine function, y , has the form $y = c \sin x + d$ and a range of $[0, 4]$. Find an expression for y , where $c > 0$.

25 A cosine function, y , has the form $y = c \cos x - d$ and a range of $[-10, 6]$. Find an expression for y , where $c > 0$.